TESS Light Curve Processing

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July 7, 2025

Overview

This section describes the basic steps for processing TESS light curves as part of the **Astroinformatics 2025-1** project. The goal was to convert observational data on exoplanet transits into clean, analyzable light curves suitable for time-series analysis and classification.

Step-by-Step Processing Workflow

1. **Data Retrieval:** TESS light curve data (FITS format) were downloaded from the MAST portal. A `.sh` script is included in the `FITS_File` directory for automated batch downloads.
2. **FITS to CSV Conversion:** FITS files were converted to `.csv` format using TOPCAT for ease of handling in Python. The resulting `.csv` files are stored in the `CSV_Files` folder.
3. **File Organization:** A script was used to generate a list of `.csv` file names and store them in a text file (`CSV_list/file_list.txt`). This allows batch processing of all available light curves.
4. **Light Curve Splitting:** Long CSV files were split into smaller segments to isolate objects into groups.
5. **Raw Light Curve Plotting:** Light curves were visualized using TOPCAT and Python. The `Light_curve/` folder contains plots exported from TOPCAT, while `LightCurve_python/` contains `.png` plots generated with `matplotlib`.
6. **Converting Light Curves to `.lc` Format:** The cleaned CSV light curves were converted to an `.lc` format compatible with time-series tools such as `LombScargle`. These are stored in the `LC_File` directory.
7. **Outlier Detection:** Using the `astropy.stats` module, statistical filters were applied to identify and, optionally, remove anomalous flux points.
8. **Time-Series Analysis:** The periodicity of the refined light curves was analyzed using the Lomb–Scargle periodogram. The periodograms and phase-folded light curves are stored in the `Statistical_Analysis` folder.

Output Summary

- `FITS_File`: Original FITS files and download script
- `CSV_Files`: Converted `.csv` files from FITS
- `CSV_list`: Script and list of filenames
- `Splitting`: Scripts and resulting split files
- `Light_curve`: Plots generated in TOPCAT

- `LightCurve_python`: `matplotlib`-generated plots
- `LC_File`: Converted `.lc` format light curves
- `Outliers`: Outlier detection code and visual outputs
- `Statistical_Analysis`: Periodograms, phased curves, statistical summaries